

CHEROKEE COUNTY AP, TX, USA

WMO: 720298

Lat: 31.869N

Lon: 95.218W

Elev: 677

StdP: 14.34

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53971

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.2	29.9	10.7	9.7	35.3	15.0	12.0	39.0	19.1	57.4	17.6	57.5	5.6	0	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.0	97.4	73.2	95.3	73.6	93.1	74.0	78.4	86.9	77.8	86.5	77.2	85.8	5.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.7	142.8	80.7	75.7	137.9	80.1	75.1	134.9	79.9	42.4	86.7	41.6	86.6	41.0	86.1	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.8	14.8	12.6	DB	19.7	99.9	4.4	2.7	16.6	101.8	14.0	103.3	11.5	104.8	8.4	106.8
			WB	17.5	79.1	4.1	1.5	14.6	80.2	12.2	81.0	10.0	81.9	7.0	82.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.7	48.4	52.2	59.3	66.1	73.3	80.2	82.4	83.3	78.4	68.3	57.8	50.3
	DBStd	14.46	10.02	10.32	9.05	6.99	5.96	3.62	3.30	3.92	4.87	7.86	9.02	9.63
	HDD50	423	152	89	28	3	0	0	0	0	0	1	31	119
	HDD65	1960	518	371	217	70	13	0	0	0	1	60	246	464
	CDD50	6532	103	150	316	485	721	907	1004	1033	852	568	265	128
	CDD65	2594	4	12	40	102	269	457	539	568	403	162	30	8
	CDH74	24863	7	39	182	605	2064	4537	5864	6576	3762	1108	105	14
	CDH80	10893	0	2	14	97	671	2007	2835	3364	1601	293	9	0
Wind	WSAvg	6.0	6.5	7.0	7.3	7.3	6.3	5.4	4.9	4.9	4.4	5.4	6.0	6.2
Precipitation	PrecAvg	42.6	3.2	3.1	3.9	4.2	4.6	3.8	2.6	2.3	3.9	3.8	3.5	3.8
	PrecMax	63.0	7.7	6.0	10.5	12.1	10.6	9.9	6.4	6.3	10.0	10.9	7.2	8.6
	PrecMin	25.0	0.3	0.8	0.9	0.8	0.7	0.4	0.6	0.1	0.3	0.0	0.4	0.8
	PrecStd	8.9	2.0	1.3	2.6	2.8	2.6	3.0	1.4	1.6	2.5	3.0	1.8	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	78.7	81.9	86.5	91.4	97.1	99.3	100.4	96.8	89.7	81.5	75.7
		MCWB	59.8	61.8	64.8	67.8	72.1	71.6	72.9	71.3	70.8	73.5	69.1	65.7
	2%	DB	71.2	74.1	78.9	82.3	88.9	94.5	96.8	98.6	93.5	86.4	77.2	72.4
		MCWB	59.5	62.8	64.5	67.0	71.6	72.8	73.0	73.0	72.7	72.1	66.6	65.2
	5%	DB	67.6	70.7	75.5	80.4	86.4	91.5	94.6	96.1	91.0	83.4	73.4	69.4
		MCWB	60.1	62.4	63.3	66.1	71.5	74.0	74.0	74.5	72.8	70.3	65.2	64.4
	10%	DB	63.9	67.9	72.5	77.5	84.0	90.0	91.5	93.5	88.6	80.8	71.6	66.0
		MCWB	58.0	61.5	61.9	65.2	71.0	74.1	74.9	74.2	72.3	68.1	65.3	61.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.2	68.6	69.4	73.7	76.4	79.2	79.0	79.3	78.9	76.6	72.3	69.9
		MCDB	69.2	73.1	74.9	79.0	83.8	87.5	87.9	87.7	86.1	82.5	76.8	72.2
	2%	WB	64.4	66.9	68.1	71.3	75.0	78.0	78.2	78.3	77.5	75.2	70.2	67.8
		MCDB	67.2	70.5	73.5	77.6	82.9	85.8	87.1	87.3	85.0	81.8	73.8	70.6
	5%	WB	63.0	64.9	66.5	69.7	73.9	77.1	77.4	77.8	76.4	73.9	68.2	65.6
		MCDB	65.8	68.7	72.0	75.7	81.7	84.8	86.4	87.1	83.6	79.9	72.1	68.9
	10%	WB	59.0	62.6	65.0	68.4	72.9	76.2	76.8	77.1	75.3	72.4	66.0	62.4
		MCDB	62.7	66.4	70.5	74.3	80.4	84.0	85.6	86.4	82.4	77.7	70.6	65.1
Mean Daily Temperature Range	5% DB	MDBR	17.1	17.2	18.4	18.8	17.8	17.3	18.4	19.0	18.2	18.4	17.2	16.4
		MCDBR	19.7	18.8	19.4	20.0	19.7	19.8	21.5	21.8	20.5	19.2	18.2	18.8
		MCWBR	12.6	11.8	9.4	8.9	6.8	5.0	4.7	4.8	5.9	7.5	10.6	13.9
	5% WB	MCDBR	16.3	16.6	16.9	16.8	17.1	16.4	17.6	18.3	17.1	16.4	16.5	17.9
		MCWBR	13.6	12.5	11.3	9.7	7.2	5.6	5.3	4.9	5.6	7.9	12.4	15.4
Clear-Sky Solar Irradiance	taub	0.318	0.337	0.357	0.399	0.432	0.446	0.457	0.456	0.417	0.359	0.349	0.325	
	taud	2.518	2.474	2.425	2.313	2.271	2.276	2.281	2.278	2.340	2.482	2.463	2.508	
	Ebn at Noon	288	292	293	283	273	268	264	263	270	281	273	279	
	Edh at Noon	27	31	35	41	43	43	43	42	38	31	29	26	
All-Sky Solar Radiation	RadAvg	884	990	1372	1702	1846	2005	2034	1909	1616	1311	956	771	
	RadStd	81	172	128	110	137	165	145	142	146	181	116	99	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149

Nomenclature: See separate page